



Classification of Business School Admission Status based on Applicant GPA and GMAT Scores

-
- Rylan Harrison
 - Jocelyn Murguia Macias
 - Kira Novitchkova-Burbank
 - Andrin Zuellig

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INTRODUCTION

Data & Objective

Overview

- Project Dataset: PSU School of Business Applications
- Motivation: Applications review takes a lot of time¹
- Goal: Classify into decision categories
(**admit**, **borderline**, and **do not admit**)

Dataset includes

- GPA Scores
- Graduate Management Admission Test (GMAT) scores²
- Admission Statuses
- 85 Applicants
 - First 5 of every admission status taken as test data
 - 70 applicants in training data

Table 1. Summary statistics of training data

Statistic	GPA	GMAT Score
Minimum	2.130	313.0
1st Quartile	2.638	419.2
Median	3.030	482.5
Mean	2.992	489.9
3rd Quartile	3.365	545.2
Maximum	3.800	693.0

Table 2. Admission status count of training data

Admission Status	Count
Admit	26
Borderline	21
Do Not Admit	23

Exploratory Data Analysis: Training Data

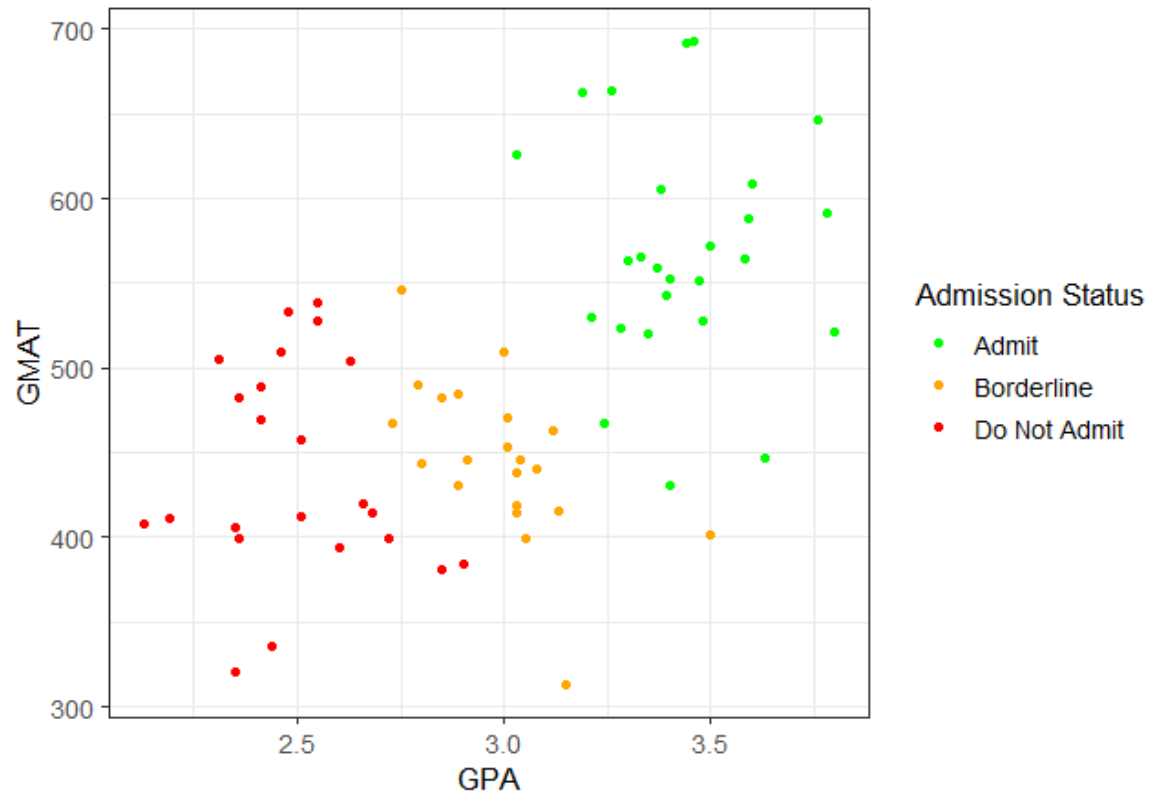


Figure 1. GMAT Scores vs GPA

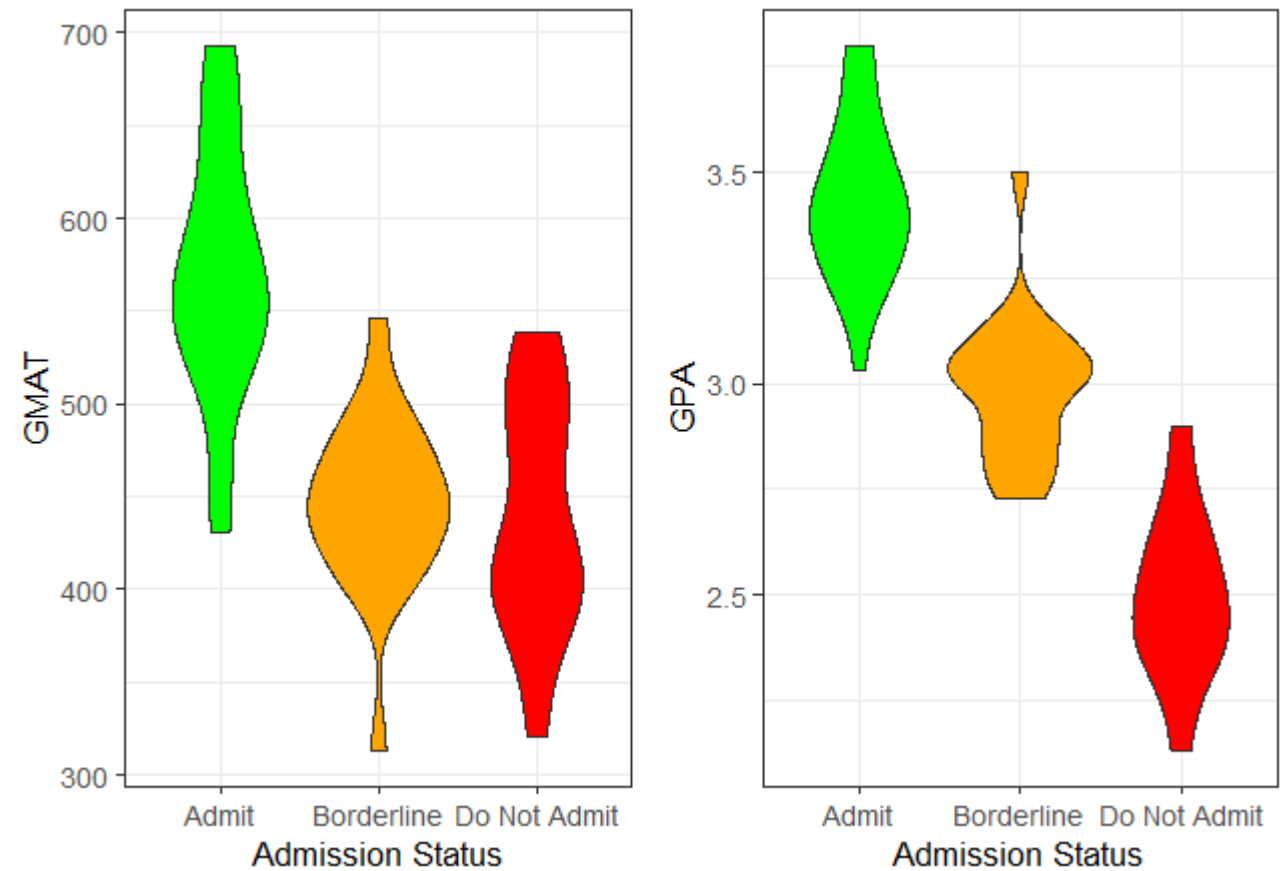
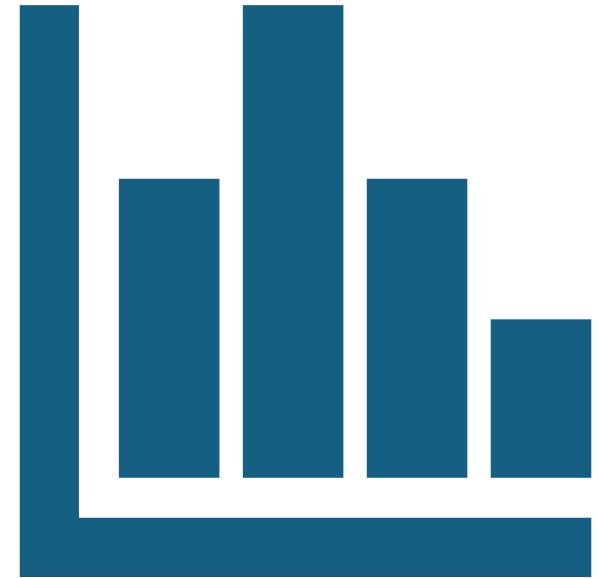


Figure 2. Distribution of GMAT Scores (left panel) and GPA (right panel)

METHODOLOGY



Overview

Predict admission statuses with three classifiers³

- K-Nearest Neighbors (KNN)
- Linear Discriminant Analysis (LDA)
- Quadratic Discriminant Analysis (QDA)

Compare class-specific and overall performance on train and test data

- Class specific: Sensitivity, Specificity
- Overall: Overall Error Rate, Mean Sensitivity and Specificity, AUC
- Visualize with decision boundaries, confusion matrices

Performance Metrics

- With caret library⁴, calculate train/test error rate, and:
 - **Sensitivity** = TP/P = fraction of “positive” classes correct
 - **Specificity** = TN/N = fraction of “negative” classes correct
- With pROC⁵ package, Hand and Till AUC: Avg. probability a class i applicant has a higher probability for class i than a class j applicant⁶
- **Intuitively**: How well does each classifier separate classes?

Classifiers: Discriminant Analysis

- **LDA** (MASS library)⁷: Model distribution of GPA, GMAT given admission status k
 - Assume normal distribution with mean μ_k and covariance Σ (estimate with train)
 - Transform probability functions for discriminant functions $\delta_k(x)$
 - Predicted class is maximizing k
- **QDA**: Same as LDA, but assume different Σ_k for each k
 - Produces quadratic discriminant functions

Classifiers: K-Nearest Neighbors

- **KNN** (caret⁴): Probability of test observation belonging to class c is proportion of class c among K -nearest neighbors
- Predicted class is c in highest proportion
- Standardized train and test to common scale
- Optimal K is 3, after computing test error for $K = 1, 2, \dots, 70$

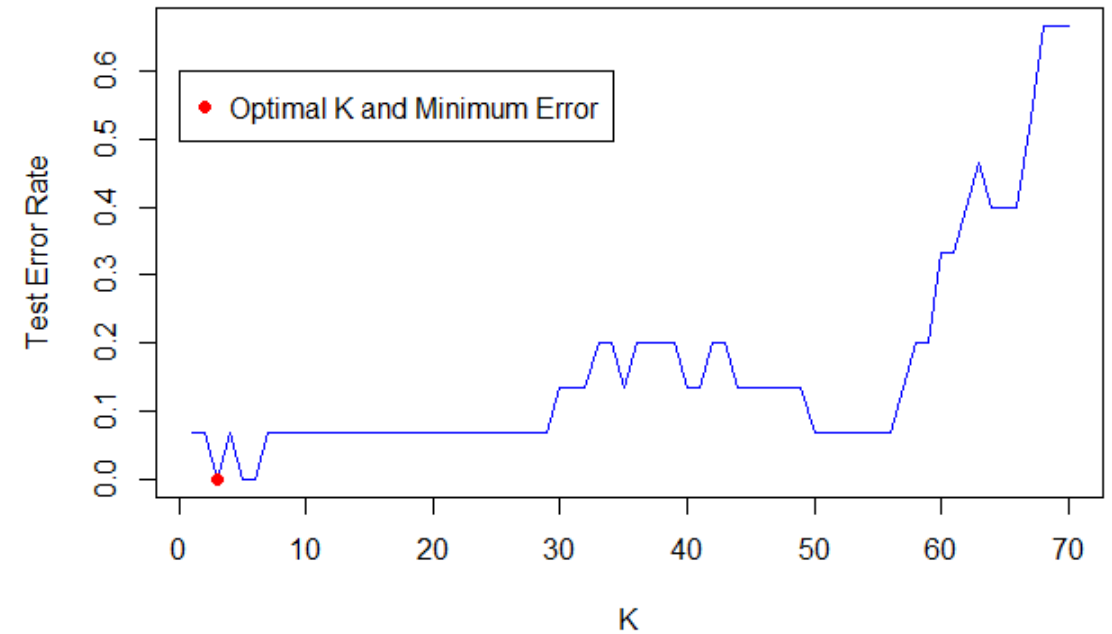
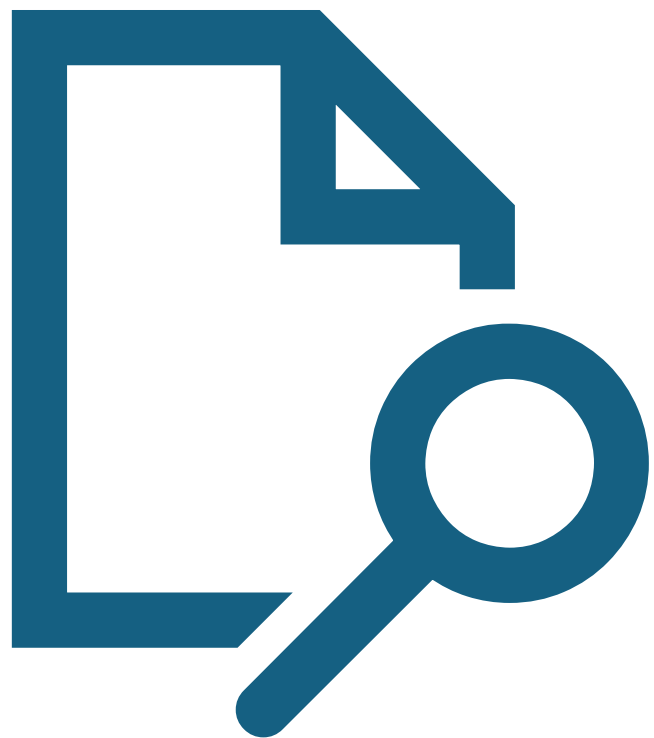


Figure 3. KNN Test Error Rates for $K = 1, 2, \dots, 70$.



RESULTS

Linear Discriminant Analysis

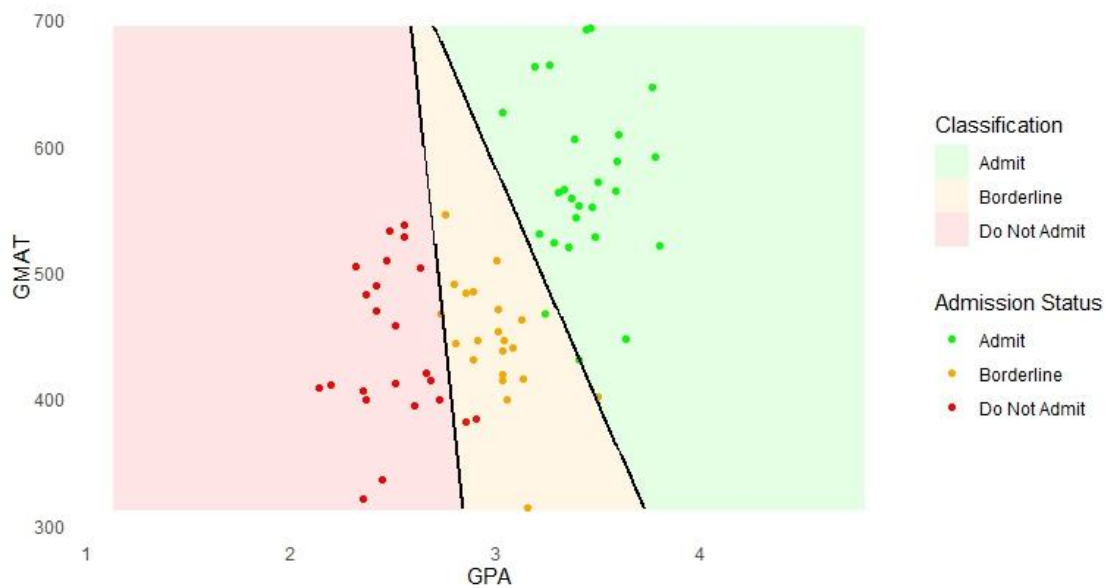


Figure 4. LDA Decision Boundary Plot on Training Data

		Predicted Class			Total
		Admit	Borderline	Do Not Admit	
True Class	Admit	24	2	0	26
	Borderline	1	19	1	21
	Do Not Admit	0	2	21	23
	Total	25	23	22	70

Table 3. Confusion Matrix of LDA on Training Data

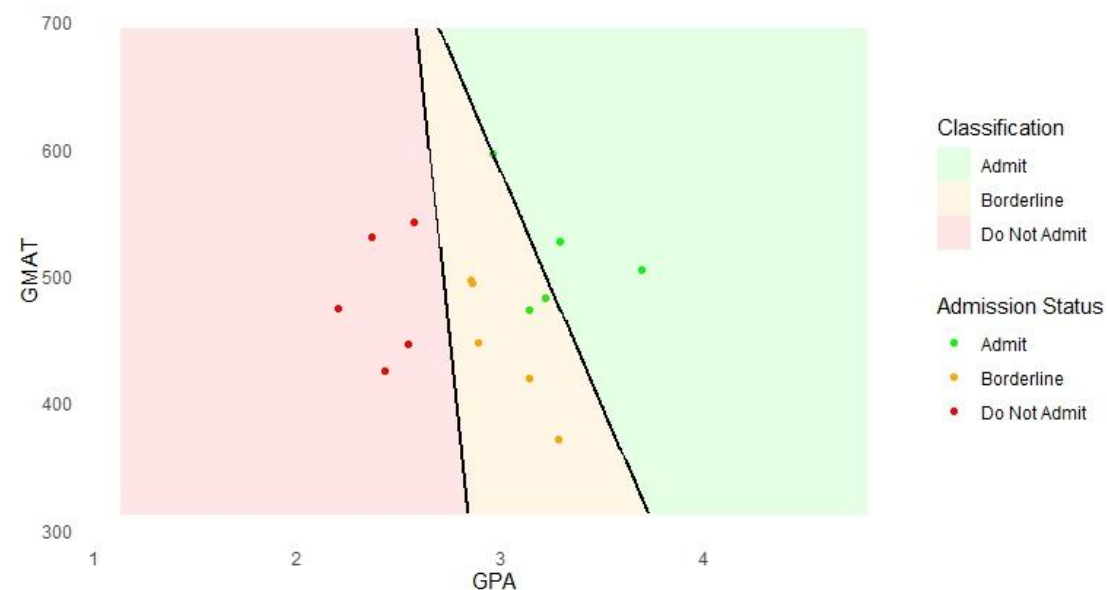


Figure 5. LDA Decision Boundary Plot on Test Data

		Predicted Class			Total
		Admit	Borderline	Do Not Admit	
True Class	Admit	2	3	0	5
	Borderline	0	5	0	5
	Do Not Admit	0	0	5	5
	Total	2	8	5	15

Table 4. Confusion Matrix of LDA on Test Data

Quadratic Discriminant Analysis

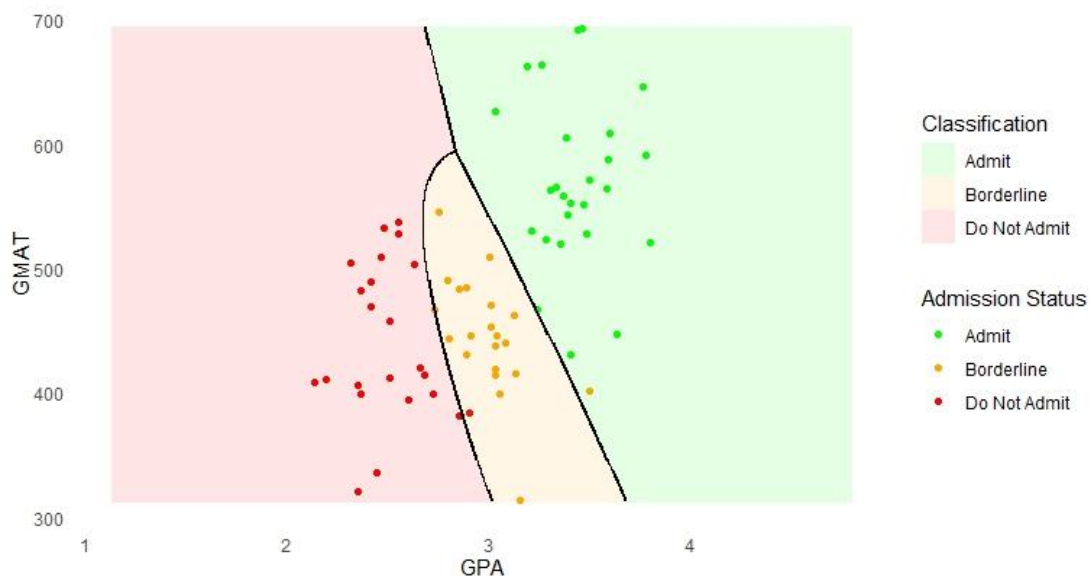


Figure 6. QDA Decision Boundary Plot on Training Data

		Predicted Class			Total
		Admit	Borderline	Do Not Admit	
True Class	Admit	26	0	0	26
	Borderline	1	20	0	21
	Do Not Admit	0	1	22	23
	Total	27	21	22	70

Table 5. Confusion Matrix of QDA on Training Data

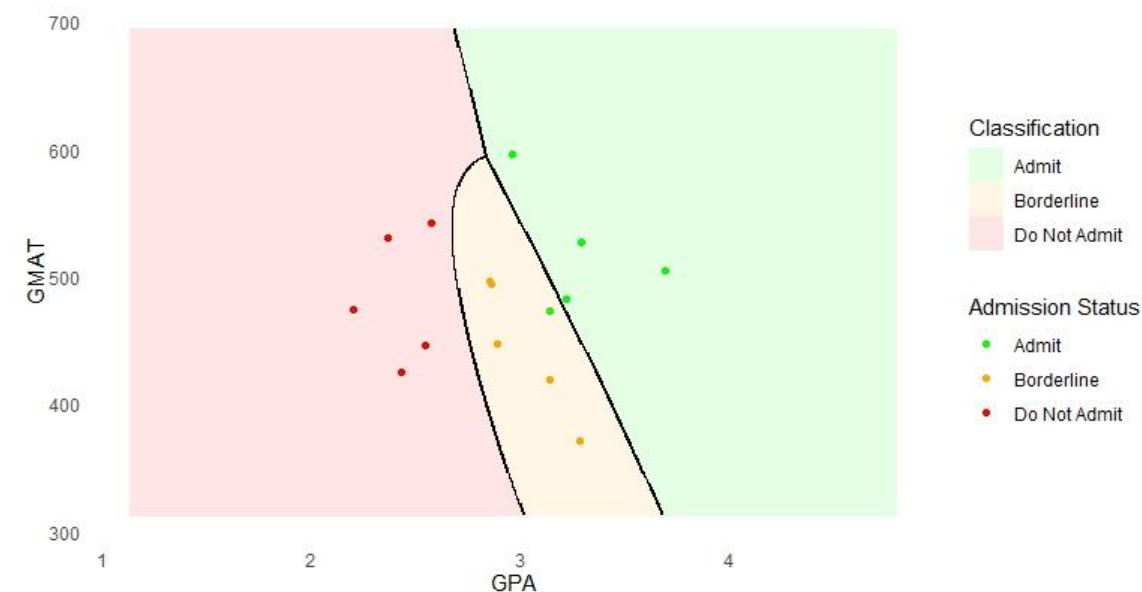


Figure 7. QDA Decision Boundary Plot on Test Data

		Predicted Class			Total
		Admit	Borderline	Do Not Admit	
True Class	Admit	4	1	0	5
	Borderline	0	5	0	5
	Do Not Admit	0	0	5	5
	Total	4	6	5	15

Table 6. Confusion Matrix of QDA on Test Data

K-Nearest Neighbors

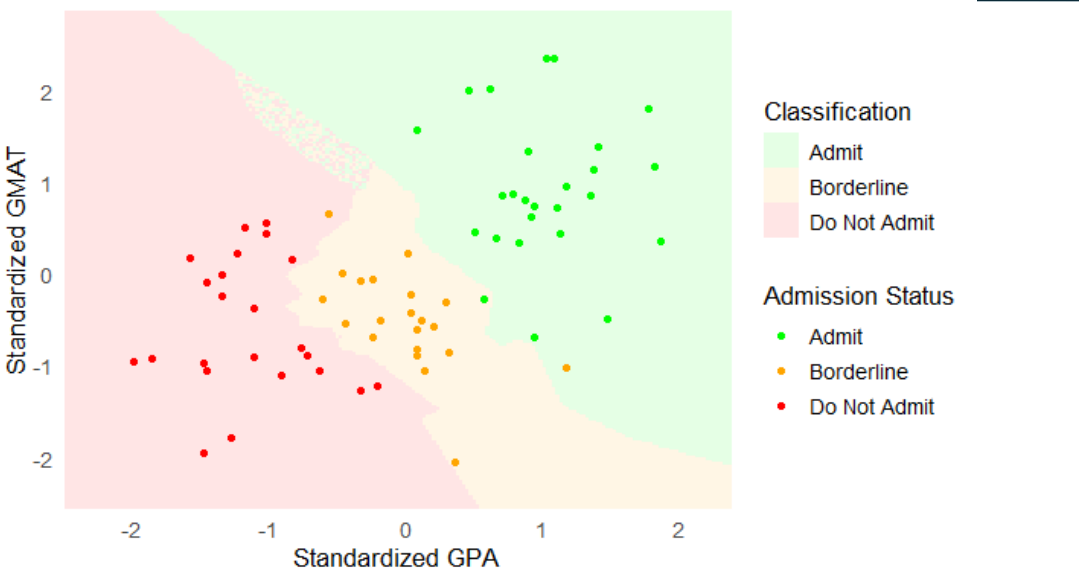


Figure 8. KNN Decision Boundary Plot on Training Data

		Predicted Class			Total
		Admit	Borderline	Do Not Admit	
True Class	Admit	25	1	0	26
	Borderline	1	19	1	21
	Do Not Admit	0	0	23	23
	Total	26	20	24	70

Table 7. Confusion Matrix of KNN on Training Data

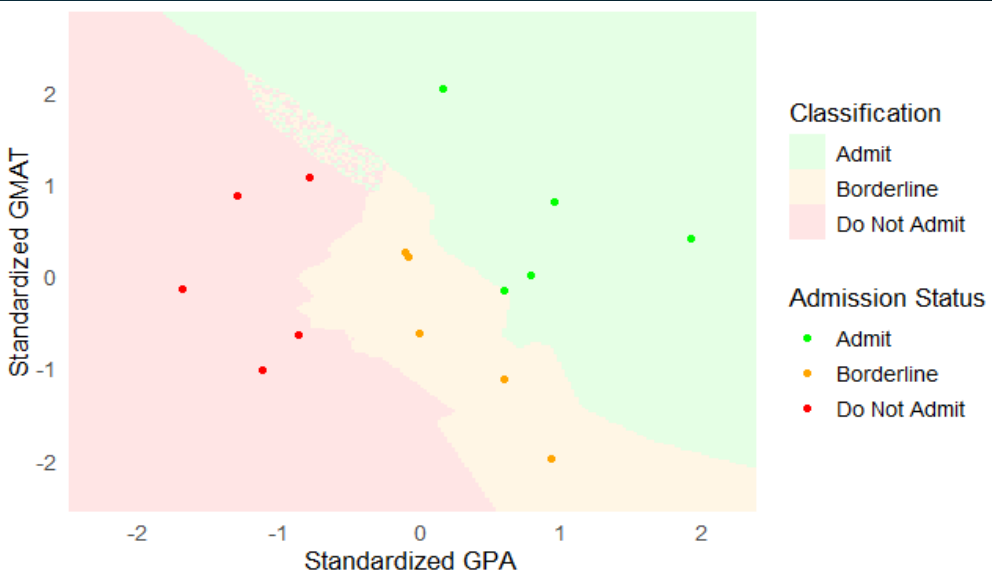


Figure 9. KNN Decision Boundary Plot on Test Data

		Predicted Class			Total
		Admit	Borderline	Do Not Admit	
True Class	Admit	5	0	0	5
	Borderline	0	5	0	5
	Do Not Admit	0	0	5	5
	Total	5	5	5	15

Table 8. Confusion Matrix of KNN on Test Data

All Classifiers: By Class

Classifier	<u>Sensitivity</u>			<u>Specificity</u>		
	Admit	Borderline	Do Not Admit	Admit	Borderline	Do Not Admit
LDA	0.923	0.905	0.913	0.977	0.918	0.979
QDA	1.000	0.952	0.957	0.977	0.980	1.000
KNN	0.962	0.905	1.000	0.977	0.980	0.979

Table 9. Sensitivity and Specificity per Class (i.e. Admit, Borderline, Do Not Admit) with Training Data

Table 10. Sensitivity and Specificity per Class (i.e. Admit, Borderline, Do Not Admit) with Test Data

Classifier	<u>Sensitivity</u>			<u>Specificity</u>		
	Admit	Borderline	Do Not Admit	Admit	Borderline	Do Not Admit
LDA	0.400	1.000	1.000	1.000	0.700	1.000
QDA	0.800	1.000	1.000	1.000	0.900	1.000
KNN	1.000	1.000	1.000	1.000	1.000	1.000

All Classifiers: Overall Performance

Classifier	Average Sensitivity	Average Specificity	Overall Error Rate	AUC
LDA	0.914	0.958	0.086	0.993
QDA	0.970	0.986	0.029	0.996
KNN	0.955	0.979	0.043	0.997

Table 11. Aggregate Performance Metrics for the Training Data

Table 12. Aggregate Performance Metrics for the Test Data

Classifier	Average Sensitivity	Average Specificity	Overall Error Rate	AUC
LDA	0.800	0.900	0.200	0.980
QDA	0.933	0.967	0.067	1.000
KNN	1.000	1.000	0.000	1.000



CONCLUSION

Observations & Recommendation

- QDA and KNN performed similar on training data, QDA slightly better OER
- KNN performed the best on the test data: only classifier with 100% accuracy
 - QDA still had AUC of 1 despite imperfect accuracy
 - → Very good at separating classes, less confident for misclassifications
- Final recommendation: KNN classifier
- Caveats: Small test set, possible overfitting (saw test data when choosing K)

References

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Thank you!

Questions?